Reviewer Pack — Polymatroid Rank Inverse Proportionality to SOS Refutation S…

Entry 75387455e1b8 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-19 14:14:45 UTC

Polymatroid Rank Inverse Proportionality to SOS Refutation Size

Entry ID: `75387455e1b8` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-19 14:14:45 UTC

1. Conjecture

Field A (mathematical branch): Polymatroid Optimization **Field B** (complexity object): SOS Refutation Size for CSPs

Statement:

> For a CSP instance with constraint matrix M, the polymatroid rank r(M) satisfies r(M) ≥ Ω(n/ρ), where ρ is the SOS refutation size. For k-CLIQUE instances, r(M) = Ω(n^{1/2}) while ρ = O(n^{1/2} log n).

Rationale (proposer's reasoning):

> Polymatroid rank captures combinatorial structure of constraints, while SOS refutation size measures algebraic complexity. Their inverse proportionality suggests hidden geometric regularity in CSP hardness.

Taxonomy category: `SOS_HIERARCHY` (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `5ac1008440d91363`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> default: >=80% seeds must support, no counterexample

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM₁	LLM₂
KARP_LIPTON	SAFE	0.95	SAFE	UNCERTAIN

4. Novelty audit

Verdict: `ADJACENT_OK` against 7 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `polymatroid rank SOS refutation size CSPs` - `sum-of-squares refutation polymatroid optimization lower bounds` - `inverse proportionality polymatroid rank SOS refutation k-CLIQUE`

Top relevant hits considered: - [http://arxiv.org/abs/2604.27336v1] Strongly Refuting Random CSP without Literals - [http://arxiv.org/abs/1701.04521v1] Sum of squares lower bounds for refuting any CSP - [http://arxiv.org/abs/2411.15363v3] The Polymatroid Representation of a Greedoid, and Associated Galois Connections - [http://arxiv.org/abs/2402.07064v6] Piecewise SOS-Convex Moment Optimization and Applications via Exact Semi-Definite Programs - [http://arxiv.org/abs/2307.07605v1] First-order Methods for Affinely Constrained Composite Non-convex Non-smooth Problems: Lower Complexity Bound and Near-o - [http://arxiv.org/abs/1401.7655v2] Inversion of the star transform - [http://arxiv.org/abs/2404.04038v1] Refutability as Recursive as Provability

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 30
    M = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]
    rank = polymatroid_rank(M)
    sos_refutation_size = sos_refutation_size(M)

    if rank is None or sos_refutation_size is None:
        return {
            "metric_name": "polymatroid_rank / sos_refutation_size",
            "metric_value": None,
            "instances_tested": 1,
            "conjecture_holds": False,
            "counterexample": "mapping_undefined"
        }

    c = 0.5
    if rank < c * n / sos_refutation_size:
        return {
            "metric_name": "polymatroid_rank / sos_refutation_size",
            "metric_value": rank / (c * n / sos_refutation_size),
            "instances_tested": 1,
```

```python
            "conjecture_holds": False,
            "counterexample": f"rank {rank} < c * n / sos_refutation_size for c={c}"
        }

    return {
        "metric_name": "polymatroid_rank / sos_refutation_size",
        "metric_value": rank / (c * n / sos_refutation_size),
        "instances_tested": 1,
        "conjecture_holds": True,
        "counterexample": ""
    }

def polymatroid_rank(M):
    # Placeholder for polymatroid rank calculation
    return None

def sos_refutation_size(M):
    # Placeholder for SOS refutation size calculation
    return None

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None) / len(re
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results if r["metric_
    support_fraction = sum(r["conjecture_holds"] for r in results if r["conjecture_holds"]) / len(

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    else:
        for r in results:
            if not r["conjecture_holds"]:
                counterexample = r["counterexample"]
                first_failing_seed = seed
                break

        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_fa
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aafdd548.py", line 66, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aafdd548.py", line 24, in run_trial
    sos_refutation_size = sos_refutation_size(M)
                          ^^^^^^^^^^^^^^^^^^^
UnboundLocalError: cannot access local variable 'sos_refutation_size' where it is not associated with
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> Test crashed due to unbound local variable error, preventing data collection | next: Fix the sos_refutation_size function implementation and re-run experiments

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	61381
2	propose	ollama_remote	qwen3:8b	0	0	61200
3	preregistration	ollama_remote	qwen3:8b	0	0	28625
4	novelty	ollama_remote	qwen3:8b	0	0	24027
5	novelty	ollama_remote	qwen3:8b	0	0	23900
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14584
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10872
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10052
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8279
10	judge	ollama_remote	qwen3:8b	0	0	18626

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 261546 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/75387455e1b8.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/75387455e1b8.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/75387455e1b8.tar.gz` (if generated)